

DAY 12 - SUMMARY

Copy Constructor

A copy constructor is a constructor that creates a new object by copying the values of another object. Java does not have a built-in copy constructor like C++, but we can create one manually.

Example

```
class Student {
    String name;
    int age;
    // Parameterized Constructor
    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }
    // Copy Constructor
    Student(Student s) {
        this.name = s.name;
        this.age = s.age;
    }
    void display() {
        System.out.println(name + " " + age);
    }

    public static void main(String[] args) {

        Student s1 = new Student("Sangeetha", 21);
        Student s2 = new Student(s1);
        s1.display();
        s2.display();
    }
}
```

Output

```
Sangeetha 21
Sangeetha 21
```

this Keyword

The this keyword refers to the current object of a class.

1. this in Parameterized Constructor

Used to differentiate instance variables and local variables having the same name.

Example

```
class Student {  
    String name;  
  
    Student(String name) {  
        this.name = name;  
    }  
  
    void display() {  
        System.out.println(name);  
    }  
  
    public static void main(String[] args) {  
        Student s = new Student("Sangeetha");  
        s.display();  
    }  
}
```

Output

Sangeetha

2. this in Default Constructor

Used to call another constructor in the same class.

Example

```
class Student {  
  
    Student() {  
        this("Sangeetha");  
    }  
  
    Student(String name) {  
        System.out.println(name);  
    }  
  
    public static void main(String[] args) {  
        Student s = new Student();  
    }  
}
```

Output

Sangeetha

3. this in Method

Used to access the current object's variables inside a method.

Example

```
class Employee {  
    String name = "Sangeetha";  
  
    void show() {  
        System.out.println(this.name);  
    }  
  
    public static void main(String[] args) {  
        Employee e = new Employee();  
        e.show();  
    }  
}
```

Output

Sangeetha